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Differences Between Relative and Absolute Speed and Metabolic Thresholds in Rugby League

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Purpose: To compare relative and absolute speed and metabolic thresholds for quantifying match output in elite rugby league. ***Methods:*** Twenty-six professional players competing in the National Rugby League were monitored with global positioning systems (GPS) across a rugby-league season. Absolute speed (moderate-intensity running [$MIR_{Th} > 3.6$ m/s] and high-intensity running [$HIR_{Th} > 5.2$ m/s]) and metabolic (>20 W/kg) thresholds were compared with individualized ventilatory (first [VT_{1IFT}] and second [VT_{2IFT}]) thresholds estimated from the 30-15 Intermittent Fitness Test ($30\text{-}15_{IFT}$), as well as the metabolic threshold associated with VT_{2IFT} (HP_{metVT2}), to examine difference in match-play demands. ***Results:*** VT_{2IFT} mean values represent 146%, 138%, 167%, and 144% increases in the HIR dose across adjustables, edge forwards, middle forwards, and outside backs, respectively. Distance covered above VT_{2IFT} was *almost certainly* greater (ES range = 0.79–1.03) than absolute thresholds across all positions. Trivial to small differences were observed between VT_{1IFT} and MIR_{Th}, while small to moderate differences were reported between HP_{metVT2} and HP_{metTh}. ***Conclusions:*** These results reveal that the speed at which players begin to run at higher intensities depends on individual capacities and attributes. As such, using absolute HIR speed thresholds underestimates the physical HIR load. Moreover, absolute MIR and high metabolic thresholds may over- or underestimate the work undertaken above these thresholds depending on the respective fitness of the individual. Therefore, using relative thresholds enables better prescription and monitoring of external training loads based on measured individual physical capacities.

Keywords: 30-15 Intermittent Fitness Test, global positioning systems, team sports

Advancements in technology have led to the extensive implementation of global positioning systems (GPS) and microtechnology in team sports to quantify movement demands. The ability to more reliably quantify and interpret these demands has led to a greater understanding of the external loads experienced by athletes during training and match play, albeit with noted limitations.[1] Ultimately, such methods provide performance staff with greater capacity to monitor athletes' training loads and prescribe field conditioning, increasing the specificity of the provided stimulus. Given the repeated high-intensity intermittent nature of team-sport match play, including in rugby league,[2–5] it is important to accurately measure the physical output of these demands. As such, high-intensity-running (HIR) distance is often considered an important measure in physical match-play output and commonly reported.[4,5] This measure has been shown to distinguish between elite and subelite levels of competition in team sports, while also separating playing positions.[4–7] Furthermore, it has been revealed that 70% of high-intensity efforts (including HIR) occur prior to pivotal match moments in rugby league, reflecting their importance during match play.[2]

In addition to measuring the distance covered at predefined speeds, the assessment of acceleration and deceleration efforts is crucial in quantifying the repeated intermittent efforts performed by players.[3,5,8] The interaction of the associated metabolic cost of acceleration and velocity has led to a new approach for analyzing time–motion data.[9] Briefly, this method calculates the metabolic cost of accelerations on flat surfaces as equal to the known metabolic cost of incline running (the equivalent angle) at a constant velocity.[10] Using this acceleration value, an instantaneous energy cost can be estimated, which, if multiplied by velocity, reveals an individual's metabolic power (W/kg). This method has been applied to team sports to report the related acceleratory demands of match play,[10–12] but little research has been done in rugby league.[3,13] Delaney et al[13] established significant differences between positional groups when examining 1- to 10-minute windows of acceleratory running, suggesting that the incorporation of acceleration-based methods is necessary in the quantification of running load in rugby league. Likewise, it is reported that when compared with HIR (>4 m/s), the distance covered over a predefined high-power metabolic threshold (HP_{metTh}) was strongly affected by position, with greater acceleration demands exhibited from middle forwards (covering 76% more distance at HP_{metTh} than HIR) than outside backs (37%).[3] However, while these studies outline the interpositional running and acceleration-based demands, they fail to take into account individual physiological capacities and may misinterpret the relative stress imposed by these efforts.

Despite the propensity to describe physical performance as an index of HIR and high metabolic power, current literature provides little agreement on the most appropriate thresholds to use. Osgnach[10] has suggested the use of an absolute high metabolic-power threshold (HP_{metTh}; >20 W/kg) reflecting a power output (W) corresponding to an oxygen consumption ($\dot{V}O_2$) of ~57 $mL \cdot kg^{-1} \cdot min^{-1}$. However, while this may be appropriate for soccer athletes (who are relatively aerobically homogeneous), there is great variability in $\dot{V}O_2$ among rugby-league and other team-sport athletes due to the variability in match-play requirements.[7,14] Similarly, there has been a wide range of speed thresholds used when reporting the locomotive movement demands of team sports, without any physiological justification for their definition.[14,15]

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It has been demonstrated that team-sport athletes greatly differ in their physiological profile, resulting in varying speeds where high-intensity running begins.[14] As such, it was recently suggested that absolute high-intensity speed thresholds should be described as physical-performance speed thresholds rather than a high-intensity speed thresholds.[14]

Owing to this, there have been recent efforts to better understand physical demands of sports using relative thresholds. Gabbett[16] analyzed youth rugby-league match play using absolute and relative thresholds, relative to a player's individual peak velocity. They reported that HIR distance, when expressed in relative thresholds, increased in slower players and decreased in faster players. Furthermore, it has been revealed that distance covered at high intensity is substantially underestimated when HIR is reported as a function of an individual's second ventilatory threshold (VT_2) (determined from gas analysis during an incremental treadmill test) compared with absolute thresholds (>5.5 m/s).[14] Similarly, when using VT_2 as a reference speed, Clarke et al[15] demonstrated that absolute thresholds (>5.0 m/s) may overestimate or underestimate HIR by up to 14% during match play. While the proposed VT_2 speed threshold appears an appropriate physiological marker in examining HIR, its application in team sports may be limited due to the impracticality of regular laboratory testing for large squads.[14] As such, it is important to examine these values using field-based tests that are easy to implement and have some physiological validity.

Recently the 30-15 Intermittent Fitness Test ($30\text{-}15_{IFT}$) has been shown to be a valid and reliable measure of prolonged high-intensity intermittent-running performance in team-sport athletes,[17–19] including in rugby league.[7,20] The $30\text{-}15_{IFT}$ is easily conducted and can test multiple athletes simultaneously, making it appealing for team-sport practitioners.[17] This test has been shown to relate well to various physiological responses such as peak oxygen uptake, as well as both the first (VT_1) and second (VT_2) ventilatory thresholds.[18] Indeed, Buchheit et al[18] revealed that the mean VT_1 and VT_2 corresponded to 68% and 87%, respectively, of the final velocity achieved during the $30\text{-}15_{IFT}$ (V_{IFT}) in team-sport athletes. Given these observations, the $30\text{-}15_{IFT}$ may provide a basis on which relative speed thresholds could be established with physiological significance. In addition, it is possible to apply the power associated with VT_2 during the $30\text{-}15_{IFT}$ to individualized high metabolic-power thresholds, giving greater insight into the acceleration demands of rugby-league match play. Therefore, the purpose of this study was to identify the differences between relative and absolute speed and high metabolic-power thresholds across rugby league match play.

Methods

Design

A longitudinal research design was adopted, where GPS match-play data were collected across a complete National Rugby League (NRL) competitive season to examine the differences in previously reported absolute speed bands[5] and metabolic-power thresholds,[10] as well as individualized relative speed and metabolic-power zones established via the $30\text{-}15_{IFT}$.

Subjects

Twenty-six professional male rugby league players (26.4 ± 3.7 y, 99.7 ± 8.3 kg) competing in the NRL participated in this study. In total, 346 individual match files were analyzed across 22 home and away and 3 finals matches, across the 2013 NRL season. Players were classified as outside backs (fullbacks, wingers, and centers; n = 8, 102 individual files), adjustables (hookers, half-backs, and five-eighths; n = 6, 77 files), edge forwards (back-rowers; n = 4, 66 files), or middle forwards (props and locks; n = 8, 101 files). Interchange players were split into the positional group in which they participated for the majority of the match. Subjects were informed of the aims and procedures of the study and gave their written informed consent prior to participation. The institutional human ethics committee approved all experimental procedures.

Methodology

Match-play data were collected using portable nondifferential GPS devices, sampling at 5 Hz and interpolated to 15 Hz (SPI HPU GPSports, Canberra, Australia). The mean ± SD number of satellites during data collection was 8.9 ± 1.5. For the purpose of validity and reliability, participants were fitted with the same GPS unit each match.[1,21] GPS data were downloaded after the completion of each match using Team AMS (GPSports, Canberra, Australia) and trimmed to include only match time (average match durations were 86 ± 7, 72 ± 18, 72 ± 19, and 41 ± 14 min for outside backs, adjustables, edge forwards, and middle forwards, respectively). Once trimmed, each file was exported for analysis with both absolute and relative thresholds applied.

Absolute thresholds were taken from previous research in rugby-league match play[5] with terminology modified for the purpose of this study. Absolute speed thresholds analyzed were moderate-intensity running (MIR_{Th}; distance at speeds >3.6 m/s) and high-intensity running (HIR_{Th}; distance at speeds >5.2 m/s). Running volumes over these speed thresholds were directly compared with speeds of the first ventilatory threshold (VT_{1IFT}; distance at speeds >68% V_{IFT}) and second ventilatory threshold (VT_{2IFT}; distance at speeds >87% V_{IFT}).[18] Distance traveled over a high metabolic-power threshold was also analyzed, with previously reported methods described by Osgnach[10] (HP_{metTh}; >20 W/kg) directly compared with the metabolic power associated with VT_{2IFT} (Equation 1) (HP_{metVT2}).

$$HP_{metVT2} = EC \times vVT_{2IFT} \times GRCF \quad (1)$$

where EC = energy cost of running at constant speed on flat compact terrain = 3.6 ($J \cdot kg^{-1} \cdot m^{-1}$), vVT_{2IFT} = relative velocity at VT_{2IFT} = 87% V_{IFT}, and GRCF = grass-running correction factor = 1.29.[10]

30-15 Intermittent Fitness Test

To measure individual's mixed aerobic/anaerobic fitness levels[7] and consequently update relative thresholds throughout the season, participants were required to complete the $30\text{-}15_{IFT}$ at 4 time points across the season. This testing was interspersed by approximately 6 weeks. If $30\text{-}15_{IFT}$ performance changed during these time points, subsequent VT_{1IFT}, VT_{2IFT}, and HP_{metVT2} thresholds were modified. The intraindividual typical error (TE) across these testing points was considered low (TE: 0.42 km/h; CV: 1.9%; unpublished data). The $30\text{-}15_{IFT}$ was performed as previously described by Buchheit,[17] on a grass field in temperatures between 18°C and 23°C. Players wore their typical training attire and football boots to ensure ecological validity. Nutritional and hydration strategies were implemented as per the club guidelines, with

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players refraining from undertaking strenuous exercise or taking stimulants for 24 hours prior to testing.

Statistical Analysis

Prior to statistical analyses, assumptions of normality were tested using the Shapiro-Wilk test, and data were log-transformed where necessary. The relationship between relative and absolute threshold data was assessed using Pearson correlation coefficients (r). The magnitude of r was classified as .1–.3 small, .3–.5 moderate, .5–.7 large, .7–.9 very large, and .9 to .99 nearly perfect.[22] After this, effect sizes (ES) and 90% confidence intervals (CI) were used to describe the magnitude of difference, interpreted as ES < 0.20 trivial, 0.21–0.6 small, 0.61–1.2 moderate, 1.21–2.0 large, and >2.1 very large.[23] Furthermore, the likelihood of the observed effect was established using a progressive magnitude-based approach, where quantitative chances of the true effect were assessed qualitatively as <1%, almost certainly not; 1% to 5%, very unlikely; 5 to 25%, unlikely; 25 to 75%, possibly; 75 to 97.5%, likely; 97.5 to 99% very likely; and >99%, almost certainly.[24] A customized spreadsheet was used to calculate confidence limits and magnitude-based inferences from P values.[24]

Results

Pearson correlation coefficients (r) between distances covered above VT_{1IFT} and MIR ($r = .94$), VT_{2IFT} and HIR ($r = .94$), and HP_{metVT2} and HP_{metTh} ($r = .93$) show a strong positive relationship between absolute and relative measures of running and metabolic loads. Raw absolute and relative threshold velocities across positional groups are outlined in Table 1. Table 2 presents the distances covered over absolute and relative speed and metabolic thresholds for these positions.

Differences (expressed as effect sizes ± 90% CI) in the distance covered above each of the relative and absolute thresholds for each positional group are displayed in Figure 1. Adjustables and edge forwards covered *most likely* same distances above VT_{1IFT} when compared with MIR_{Th} (ES = 0.01; ± 0.20 and ES = 0.05; ± 0.29, respectively), while outside backs and middle forwards completed a *likely* and *very likely* greater volume of VT_{1IFT} than MIR_{Th} (ES = 0.21; ± 0.24 and ES = 0.34; ± 0.23, respectively). Moderate differences were witnessed between VT_{2IFT} and HIR_{Th} for all positional groups. The VT_{2IFT} mean values represent 146%, 144%, 138%, and 167% increases in HIR (compared with HIR_{Th}) across adjustables, outside backs, edge forwards, and middle forwards, respectively.

The relationship between VT_{2IFT} raw velocity and the differences in distance covered above VT_{2IFT} and HIR is presented in Figure 2, reflecting a tendency for players with a lower VT_{2IFT} velocity (and therefore level of fitness) to have a greater underestimation of high-intensity running when using absolute values. Relative high metabolic-power thresholds were *likely* to *almost certainly* lower than absolute HP_{metTh} across positions (ES range = 0.24–0.63). However, some athletes completed a greater volume of running above HP_{metVT2} than HP_{metTh} depending on the level of fitness of the individual.

Table 1 Velocity (Mean ± SD) for Relative and Absolute Speed and Metabolic Thresholds Across Positional Groups

Running threshold	Adjustables	Outside backs	Edge forwards	Middle forwards	Squad
VT_{1IFT} (m/s)	3.7 ± 0.1	3.6 ± 0.2	3.7 ± 0.1	3.5 ± 0.1	3.6 ± 0.2
MIR_{Th} (m/s)	3.6	3.6	3.6	3.6	3.6
VT_{2IFT} (m/s)	4.7 ± 0.2	4.5 ± 0.3	4.7 ± 0.1	4.5 ± 0.1	4.6 ± 0.2
HIR_{Th} (m/s)	5.2	5.2	5.2	5.2	5.2
HP_{metVT2} (W/kg)	22.0 ± 0.7	21.1 ± 1.5	21.7 ± 0.5	21.0 ± 0.5	21.4 ± 1.0
HP_{metTh} (W/kg)	20	20	20	20	20

Abbreviations: VT_{1IFT}, first ventilatory threshold (>68% V_{IFT}); MIR_{Th}, absolute moderate-intensity-running threshold (>3.6 m/s); VT_{2IFT}, second ventilatory threshold (>87% V_{IFT}); HIR_{Th}, absolute high-intensity-running distance (>5.2 m/s); HP_{metTh}, absolute high metabolic-power threshold (>20 W/kg); HP_{metVT2}, relative high metabolic-power threshold (power associated with VT_{2IFT}).

Table 2 Distances (Mean ± SD) Covered Across Relative and Absolute Speed and Metabolic Thresholds Across Positional Groups

Running threshold	Adjustables	Outside backs	Edge forwards	Middle forwards	Squad
VT_{1IFT} (m)	1110 ± 339	1520 ± 211[a]	1301 ± 103	836 ± 184[a]	1192 ± 358
MIR_{Th} (m)	1108 ± 365	1415 ± 208	1319 ± 103	730 ± 176	1133 ± 367
VT_{2IFT} (m)	387 ± 132[b]	809 ± 161[b]	572 ± 76[b]	316 ± 65[b]	526 ± 240
HIR_{Th} (m)	270 ± 83	535 ± 89	413 ± 76	187 ± 35	351 ± 161
HP_{metVT2} (m)	1137 ± 324[c]	1377 ± 189[c]	1296 ± 109[d]	797 ± 175[c]	1144 ± 321
HP_{metTh} (m)	1315 ± 373	1468 ± 216	1486 ± 118	851 ± 204	1264 ± 365

Abbreviations: VT_{1IFT}, first ventilatory threshold (>68% V_{IFT}); MIR_{Th}, absolute moderate-intensity-running threshold (>3.6 m/s); VT_{2IFT}, second ventilatory threshold (>87% V_{IFT}); HIR_{Th}, absolute high-intensity-running distance (>5.2 m/s); HP_{metTh}, absolute high metabolic-power threshold (>20 W/kg); HP_{metVT2}, relative high metabolic-power threshold (power associated with VT_{2IFT}).

Positional effect sizes between thresholds: [a] small effect with MIR_{Th}; [b] moderate effect with HIR_{Th}; [c] small effect with HP_{metTh}; [d] moderate effect with HP_{metTh}.

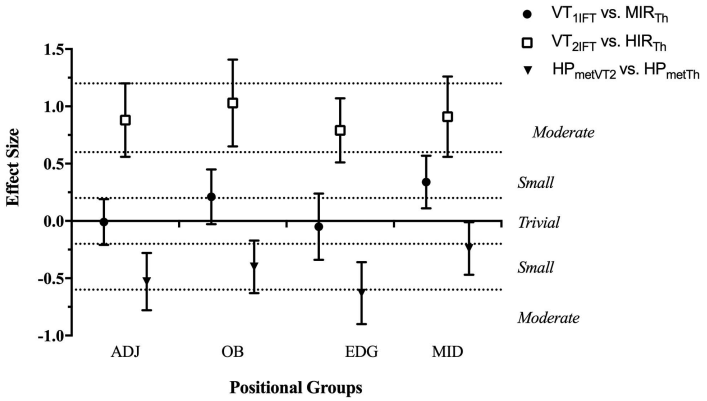

Figure 1 — Magnitude of differences in running and high metabolic distance between relative and absolute thresholds across positional groups. Values are standardized effects (effect size ± 90% CI). ADJ indicates adjustables; OB, outside backs; EDG, edge forwards; MID, middle forwards; VT_{1IFT}, first ventilatory threshold (>68% V_{IFT}); MIR_{Th}, absolute moderate-intensity-running threshold (>3.6 m/s); VT_{2IFT}, second ventilatory threshold (>87% V_{IFT}); HIR_{Th}, absolute high-intensity-running distance (>5.2 m/s); HP_{metTh}, absolute high metabolic-power threshold (>20 W/kg); HP_{metVT2}, relative high metabolic-power threshold (power associated with VT_{2IFT}).

Discussion

The purpose of this study was to identify differences between relative and absolute speed and metabolic-power thresholds in quantifying the match-play output of elite rugby league players. The initial finding in the current study demonstrates that while strong correlations exist between distances covered at relative and absolute speed thresholds, substantial differences in match-play output were evident when examined across positional groups. This investigation confirmed that HIR should be considered relative to an individual, as the running "load" of match play greatly depends on an individual's physiological capacities. Similarly, the use of estimated VT_2 as a physiological marker for high metabolic-power thresholds may provide a better reference for individual high metabolic running than absolute methods in team sports.

The present findings demonstrate that there is an almost perfect correlation between distance covered above relative and absolute speed thresholds ($r = .93$–$.94$). Additionally, while only small differences existed for distance covered above VT_{1IFT} compared with MIR_{Th}, greater (moderate) differences were observed when comparing distances above VT_{2IFT} with HIR_{Th}. These findings suggest that absolute speed thresholds may underestimate HIR in rugby-league match play, particularly for players with lower levels of fitness, mirroring results previously reported in soccer.[14] For example, Abt and Lovell[14] reported substantially greater distances in HIR (167%) during match play using VT_2 as their speed threshold, compared with an absolute HIR threshold (>5.5 m/s). The importance of HIR and repeated high-intensity efforts in rugby-league match play has previously been outlined.[2,5,6] Indeed, HIR and repeated high-intensity efforts have been shown to differentiate team success[25,26] and commonly occur around pivotal match moments.[2,6] Due to the importance of HIR and repeated high-intensity efforts to performance, there has been an augmented shift to high-intensity interval training to develop these capacities,[27] but underestimation or misrepresentation of the physical HIR load may lead to poor monitoring and prescription of training load. Furthermore, it has been proposed that an individual's ventilatory threshold (VT_2) provides a more sensitive measure of training-induced changes than typical measures of maximal aerobic capacity.[28,29] Hence, the use of relative thresholds appears desirable for appropriate player monitoring.

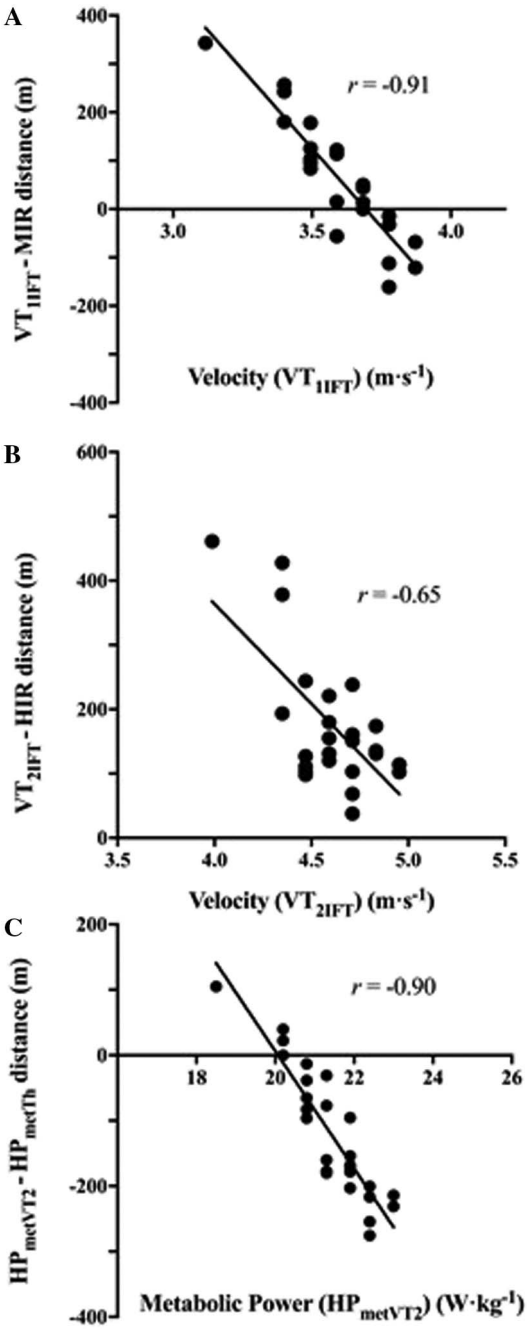

Figure 2 — (A) VT_{1IFT} raw velocity and the differences in distance covered above VT_{1IFT} versus MIR. (B) VT_{2IFT} raw velocity and the differences in distance covered above VT_{2IFT} versus HIR. (C) HP_{metVT2} raw power and the differences in distance covered above HP_{metVT2} and HP_{metTh}. VT_{1IFT} indicates first ventilatory threshold (>68% V_{IFT}); MIR_{Th}, absolute moderate-intensity-running threshold (>3.6 m/s); VT_{2IFT}, second ventilatory threshold (>87% V_{IFT}); HIR_{Th}, absolute high-intensity-running distance (>5.2 m/s); HP_{metTh}, absolute high metabolic-power threshold (>20 W/kg); HP_{metVT2}, relative high metabolic-power threshold (power associated with VT_{2IFT}).

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The current investigation observed positional differences for distances run above VT_{2IFT} in comparison with distance run above HIR_{Th}. Due to the variations in rugby-league positional match-play demands, there is little homogeneity across players' physical, physiological, and anthropometrical characteristics.[30] For instance, middle forwards are required to carry greater body mass and possess explosive qualities to increase their momentum in collisions.[31,32] In contrast, adjustables are often smaller and require greater prolonged high-intensity intermittent-running ability due to the increased running demands of their position during match play.[7] When examining the differences in distance covered above VT_{2IFT} and HIR_{Th}, all playing positions had an *almost certainly* greater output (ES range = 0.79–1.03) when HIR was considered relative to their level of fitness, with the greatest magnitude observed among middle forwards and outside backs. This is reflected in the lower V_{IFT} of middle forwards and outside backs in the current study, which mirrors previous findings in rugby league.[7] Indeed, it has been demonstrated that these 2 positional groups contribute greater anaerobic qualities during the 30-15_{IFT} than other positions, potentially limiting their performance in this test.[7] Due to the heterogeneity in physical characteristics across positions, the relative HIR velocity substantially differs between positions in rugby league. As such, when examining HIR across positions, the use of relative thresholds (eg, VT_{2IFT}) may provide a better representation of the relative physical HIR load undertaken.

It has been suggested that HIR above VT_2 during match play does not account for the entire physical running load, as the distance and/or time spent at moderate intensities is not included.[14] The current findings indicate that absolute MIR thresholds may not only underestimate the MIR loads in players with lower fitness levels but also overestimate these loads for players with higher levels of fitness. Adjustables and edge forwards covered substantially less distance above VT_{1IFT} than with MIR_{Th}. Conversely, outside backs and middle forwards completed a greater volume of VT_{1IFT} than MIR_{Th}. The small practical differences witnessed in middle forwards and outside backs are likely due to the lower V_{IFT} of these positions, resulting in greater differences between VT_{1IFT} and MIR_{Th} velocities. Furthermore, this study has demonstrated that when applying absolute MIR thresholds, individuals with high levels of fitness may have a tendency to overestimate the running undertaken above this intensity. Given these findings it appears important that relative speed thresholds also take into account the running load undertaken at moderate intensities, as this too is sensitive to individual match-play demands.

In addition to speed-based thresholds, this investigation examined the differences between high metabolic-power thresholds. HP_{metVT2} provides an estimated physiological threshold that offers a relative value across individuals with heterogeneous physiological capacities, particularly for those with lower fitness levels. The present findings demonstrate that a strong positive correlation was observed ($r = .94$) between distance run above HP_{metVT2} and HP_{metTh}. Similar to the differences witnessed between running thresholds at moderate intensities, running undertaken above absolute high metabolic-power thresholds may underestimate and overestimate running load depending on the fitness of the individual. The current study showed a main effect in distance traveled above relative and absolute high metabolic-power thresholds across positional groups. Edge forwards undertook moderately less distance above HP_{metVT2} than HP_{metTh}, while small lower differences were reported across all other positions. Despite adjustables' recording a higher relative high metabolic threshold than any other position (22.0 ± 0.7 W/kg), only small differences in high metabolic running distances were observed between thresholds. It has been previously demonstrated that edge forwards and adjustables undertake a greater metabolic-power output during match play than other positions when using HP_{metTh}.[13] As adjustables and edge forwards have higher relative metabolic-power thresholds, this may contribute in part to the previously reported greater metabolic-power outputs when using absolute thresholds than other positions during match play. Furthermore, while it has been demonstrated that HP_{metTh} may overestimate the high metabolic running load of these positions, it appears that edge forwards are most greatly affected. In addition, the current study demonstrates that for players with lower levels of fitness, absolute thresholds may underestimate this high metabolic running volume. Consequently, the use of estimated VT_2 as an individual physiological marker for high metabolic-power output may provide a better reference for individual high metabolic load in team sports.

Collectively, these findings confirm previous reports in soccer that have shown that the HIR load is underestimated using current absolute methods, which is directly affected by levels of fitness.[14] Additionally, the current study demonstrates that the differences between absolute and relative HIR load vary substantially across individuals during match play. It has been suggested that VT_2 is a more appropriate marker to implement as a relative HIR threshold due to its sensitivity in measuring training-induced physiological changes.[14,28,29] However, given the laboratory nature of incremental treadmill testing to examine ventilatory thresholds, it may be difficult to calculate these physiological functions in large team-sport squads. The current study presents the 30-15_{IFT} as a practical field-based test to implement relative thresholds for match demands due to the strong relationship between V_{IFT} and both VT_1 and VT_2.[18] Despite this, it must be noted that a limitation of this application is this assumed relationship between the 30-15_{IFT} and these thresholds, which may vary between individuals. However, while this methodology only provides estimates of these physiological markers, we recommend the use of 30-15_{IFT}-derived relative thresholds due to the practicality of this measure and its ability to be implemented across large groups. Additionally, it is important to recognize that to increase the accuracy of these thresholds, repeated testing is required across the preparation and competition phase. While the 30-15_{IFT} was performed at 4 time points across the competition season to modify relative thresholds, further research should aim at determining how variable the 30-15_{IFT} is across a competitive phase and whether this was sufficient.

Practical Applications

- VT_{2IFT} presents a relative HIR threshold that can be applied across a large squad in team sports. It is proposed that this estimated threshold may provide a more practical value than laboratory-based VT_2 testing.
- Current absolute HIR thresholds (>5.5 m/s) underestimate the physical running load undertaken by athletes during match play, so VT_{2IFT} may be a greater indicator of this running load, particularly for those with lower fitness levels.
- The relative nature of this "running load" may provide greater insight into the volume of load performed and consequently be used by performance staff to more precisely prescribe and monitor training sessions and recovery strategies.

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Conclusion

The present study demonstrated that considerable differences existed in match-play output across positions when relative threshold speeds (derived from the 30-15$_{IFT}$) were compared with absolute methods. Similar to previous findings in soccer,[14] it is apparent that the speed at which players begin HIR depends on individual capacities and attributes. As such, current absolute methods underestimate the HIR undertaken by rugby-league players during match play. Furthermore, this study found that absolute MIR may not only underestimate the MIR loads in players with lower fitness levels but also overestimate these loads for players with higher levels of fitness, confirming the suggestions that this needs to be investigated in team sports. Likewise, it is apparent that absolute high metabolic-power thresholds may underestimate and overestimate this load depending on the fitness of the individual compared with VT$_{2IFT}$-based high metabolic-power thresholds. Taken together, these findings will affect our understanding of the HIR running "load" to which athletes are exposed during both training and match play. These findings may have great effect on the periodization of training, recovery protocols, and prescription of high-intensity interval training for performance staff.

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